

WEBSTER NOLF, MD, USA

WMO: 725514

Lat: 38.142N

Lon: 76.429W

Elev: 6

StdP: 101.25

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 03757

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.5	-6.3	-19.0	0.7	-6.9	-16.0	0.9	-4.3	10.6	0.7	9.3	0.7	3.0	300	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	33.8	25.1	32.4	24.4	31.1	23.8	26.4	31.6	25.7	30.4	25.1	29.5	3.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.0	20.1	29.3	24.4	19.3	28.6	23.8	18.6	27.9	82.1	31.7	79.0	30.4	76.4	29.6	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.4	6.5	-12.2	35.5	3.3	1.8	-14.6	36.8	-16.5	37.8	-18.3	38.9	-20.7	40.2		
(5)			-13.1	27.4	3.1	0.9	-15.3	28.0	-17.0	28.5	-18.8	29.0	-21.0	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	2.6	4.1	8.0	14.1	19.7	24.1	26.2	24.9	21.9	16.2	9.8	5.7
	DBStd	9.31	5.51	5.51	5.41	4.72	4.19	3.07	2.66	2.61	3.44	4.59	4.38	4.83
	HDD10.0	746	238	177	105	15	0	0	0	0	0	4	56	151
	HDD18.3	2137	487	397	321	143	36	1	0	0	8	96	257	391
	CDD10.0	2516	9	13	44	137	302	424	503	462	357	197	49	19
	CDD18.3	866	0	0	1	15	79	175	245	204	114	30	2	1
	CDH23.3	7550	0	0	12	127	546	1511	2523	1835	832	162	2	0
	CDH26.7	2511	0	0	1	29	149	509	978	578	233	33	0	0
Wind	WSAvg	2.7	3.0	3.1	3.2	3.2	2.8	2.6	2.4	2.1	2.3	2.7	2.7	2.7
Precipitation	PrecAvg	1168	82	72	99	90	92	112	115	126	106	100	77	93
	PrecMax	1667	183	187	258	199	224	264	262	258	274	204	198	202
	PrecMin	899	38	13	1	48	29	40	38	37	22	17	9	39
	PrecStd	175	33	39	48	32	50	53	53	52	65	55	48	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.9	20.8	25.1	30.3	32.3	34.8	36.5	34.1	33.1	29.7	22.3	20.9
		MCWB	13.8	15.2	16.6	19.6	23.8	24.5	25.9	25.7	24.1	22.6	17.7	17.7
	2%	DB	16.3	18.0	21.7	26.8	30.1	32.8	34.3	32.5	31.1	27.1	20.5	18.5
		MCWB	12.9	13.1	14.8	17.9	22.0	23.7	25.5	25.1	23.4	21.2	16.2	15.5
	5%	DB	13.9	15.2	19.3	24.3	28.3	31.2	32.8	31.1	29.3	25.0	18.4	16.2
		MCWB	11.1	11.6	13.5	17.3	21.4	23.4	25.1	24.2	22.5	20.4	14.9	14.0
	10%	DB	11.4	12.6	16.5	21.9	26.5	29.6	31.2	29.8	27.6	23.2	16.7	13.3
		MCWB	8.3	8.7	11.9	15.9	20.2	22.6	24.3	23.8	21.9	18.8	13.6	10.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.3	17.1	17.9	21.7	24.6	26.2	27.7	27.2	25.8	24.2	19.2	18.8
		MCDB	17.0	20.0	22.8	24.7	30.4	32.3	33.2	32.4	30.3	27.2	21.6	20.3
	2%	WB	13.4	14.1	16.2	19.5	23.3	25.2	26.7	26.2	24.8	22.6	17.4	15.9
		MCDB	15.8	16.5	20.7	24.2	28.2	30.4	32.0	30.9	28.9	25.7	19.2	17.8
	5%	WB	11.5	11.8	14.5	18.1	22.3	24.6	26.0	25.5	24.0	21.2	15.6	14.0
		MCDB	13.8	14.9	18.3	23.2	26.6	29.3	30.9	29.6	27.3	24.3	17.8	15.8
	10%	WB	8.6	9.1	12.5	16.8	21.4	23.9	25.3	24.7	23.1	19.9	14.0	11.4
		MCDB	11.0	12.2	15.8	21.1	25.3	28.2	29.9	28.5	26.3	22.7	16.4	13.0
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.9	9.5	10.3	9.3	9.1	8.7	8.6	8.8	9.3	8.9	8.3
		MCDBR	11.0	12.0	12.8	12.9	10.6	10.4	9.8	9.1	10.2	9.9	10.4	10.0
		MCWBR	9.5	8.9	7.7	6.2	5.1	4.3	3.8	3.9	4.7	5.2	7.1	8.8
	5% WB	MCDBR	10.2	11.2	11.7	11.0	9.2	9.4	8.6	8.0	7.9	8.7	8.6	9.5
MCWBR		9.5	9.0	8.1	6.3	4.7	4.2	3.9	3.9	4.3	5.2	6.9	9.0	
Clear-Sky Solar Irradiance	taub	0.310	0.319	0.353	0.393	0.434	0.484	0.516	0.508	0.410	0.359	0.326	0.312	
	taud	2.518	2.485	2.432	2.336	2.257	2.162	2.088	2.095	2.347	2.490	2.542	2.550	
	Ebn at Noon	876	912	908	886	852	806	776	772	837	854	850	849	
	Edh at Noon	78	91	106	123	136	150	160	155	114	90	76	71	
All-Sky Solar Radiation	RadAvg	2.22	3.03	4.04	5.11	5.81	6.37	6.16	5.52	4.47	3.42	2.51	1.96	
	RadStd	0.17	0.32	0.34	0.36	0.63	0.47	0.29	0.30	0.44	0.38	0.22	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (79 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140	+86

Nomenclature: See separate page